4 Convergence analysis and finite-trajectory projection bounds

Finite-pool descent accounts for continuous-dictionary coverage, score evaluation, radial stationarity, and coefficient correction. Local Hessian transfer connects nonlinear selection with weak OGA in a frozen Hilbert metric. Projection estimates describe the nested spaces generated by the algorithm, and the quartic identity converts their approximation errors into energy bounds.

Throughout, m counts accepted independent directions, m_0 is the starting index, and $j = m - m_0$. We use $\widehat{G}_m$ for the computed state, $G_m^\sharp$ for the exact active-space energy minimizer, and z_m for the frozen orthogonal projection. All derivatives below act on the specified feasible linear space, so that $\mathcal{E}'(u^*) = 0$ there. Affine constraints are reduced to this setting by a fixed feasible lift.

4.1 Finite-pool descent and the residual-comparison condition

Let X be a real Banach space and $\mathcal{E} : X \to (-\infty, +\infty]$ a proper convex energy with minimizer u^*. Assume that $\mathcal{E}$ is finite and continuously differentiable on an open neighborhood containing the states and dictionary segments used below. The symmetric dictionary $\mathcal{D}_X$ consists of X-unit vectors. Set

$$\widehat{\delta}_m = \mathcal{E}(\widehat{G}_m) - \mathcal{E}(u^*), \qquad S_m^X = \sup_{g \in \mathcal{D}_X} \left| \left\langle \mathcal{E}'(\widehat{G}_m), g \right\rangle \right|.$$

The exact finite-pool score is $s_m(g) = |\langle \mathcal{E}'(\widehat{G}_m), g\rangle|$. The available computed scores satisfy

$$|\widetilde{s}_m(g) - s_m(g)| \le \xi_m \quad (g \in \mathcal{P}_m), \qquad \widetilde{s}_m(g_{m+1}) \ge t_m^{\mathrm{pool}} \max_{g \in \mathcal{P}_m} \widetilde{s}_m(g), \tag{1}$$

where $\mathcal{P}_m \subset \mathcal{D}_X$ is finite and $0 < t_m^{\mathrm{pool}} \le 1$. When $S_m^X > 0$, define its continuous coverage by

$$\tau_m = \frac{\max_{g \in \mathcal{P}_m} s_m(g)}{S_m^X} \in [0, 1]. \tag{2}$$

Set $\tau_m = 1$ when $S_m^X = 0$. The selection weakness t_m^{pool} and the coverage τ_m are different quantities.

Assume that a common $L > 0$ satisfies

$$\mathcal{E}(\widehat{G}_m + \lambda g) \le \mathcal{E}(\widehat{G}_m) + \lambda \langle \mathcal{E}'(\widehat{G}_m), g\rangle + \tfrac{L}{2}\lambda^2, \qquad |\lambda| \le 1, \quad g \in \mathcal{D}_X. \tag{3}$$

The expanded space contains $\widehat{G}_m + \lambda g_{m+1}$, and the correction satisfies

$$\mathcal{E}(\widehat{G}_{m+1}) \le \inf_{v \in \widehat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\mathrm{opt}}, \qquad \varepsilon_m^{\mathrm{opt}} \ge 0. \tag{4}$$

The active-space energy comparison relates this correction estimate to the numerical objective.

Proposition 4.1 (Descent with a resolved score)**.** *Let $a_m \ge 0$ be any valid lower bound for $s_m(g_{m+1})$. Under* (3)–(4)*,*

$$\widehat{\delta}_{m+1} \le \widehat{\delta}_m - H_L(a_m) + \varepsilon_m^{\mathrm{opt}}. \tag{5}$$

Here H_L is the directional-decrease function defined in the finite-pool descent estimates. In particular, (1) *gives the valid lower bound*

$$a_m = \left[t_m^{\mathrm{pool}} \tau_m S_m^X - (1 + t_m^{\mathrm{pool}}) \xi_m \right]_+ . \tag{6}$$

Proof. The triangle inequality gives $s_m(g_{m+1}) \ge t_m^{\mathrm{pool}} \max_{\mathcal{P}_m} s_m - (1 + t_m^{\mathrm{pool}})\xi_m$. Test the correction with a signed dictionary step in the true descent direction of length $\min\{1, a_m/L\}$. Both signs lie in the expanded linear space. Minimizing the quadratic upper bound on that segment gives $H_L(a_m)$ and proves (5). □

The radial stationarity error enters when an atomic source is used to bound S_m^X. Write $K_1(\mathcal{D})$ for the closed absolutely convex hull of a bounded dictionary. If $u^* \in B_{\mathrm{src}} K_1(\mathcal{D}_X)$ with $B_{\mathrm{src}} > 0$ and $|\langle \mathcal{E}'(\widehat{G}_m), \widehat{G}_m \rangle| \le \eta_m^{\mathrm{stat}}$, convexity implies

$$\widehat{\delta}_m \le \eta_m^{\mathrm{stat}} + B_{\mathrm{src}} S_m^X. \tag{7}$$

Consequently one may instead use

$$a_m = \left[t_m^{\mathrm{pool}} \tau_m \frac{\widehat{\delta}_m - \eta_m^{\mathrm{stat}}}{B_{\mathrm{src}}} - (1 + t_m^{\mathrm{pool}}) \xi_m \right]_+ .$$

This source-dependent estimate expresses the selected-score lower bound in terms of the energy gap and radial stationarity. In the error framework of [3, Section 3, Theorem 3.2 and Lemma 6.3], energy correction and radial biorthogonality are separate errors. Their power-smoothness rate assumes uniform positive continuous weakness and prescribed decay of those errors. Here they correspond to $\varepsilon_m^{\mathrm{opt}}$ and η_m^{stat}, with the latter at the preceding state in (7). Pool coverage and score evaluation determine the effective selection strength through (6).

Lemma 4.2 (A nonautonomous scalar bound)**.** *Suppose $x_m \ge 0$ and $x_{m+1} \le x_m - b_m x_m^q$ for $q > 1$ and $b_m \ge 0$. For $x_0 > 0$,*

$$x_n \le \left[x_0^{1-q} + (q-1) \sum_{m=0}^{n-1} b_m \right]^{-1/(q-1)} . \tag{8}$$

If $x_0 = 0$, the sequence vanishes. In particular, $b_m \ge b(m+1)^{-s}$, $b > 0$, $0 \le s < 1$, implies $x_n = O((n+1)^{-(1-s)/(q-1)})$.

Proof. As long as $x_{m+1} > 0$, convexity gives $x_{m+1}^{1-q} - x_m^{1-q} \ge (q-1) x_m^{-q} (x_m - x_{m+1}) \ge (q-1) b_m$. Sum. Once a term vanishes, nonnegativity and the recurrence force all later terms to vanish. The power estimate follows by comparison of the sum with an integral. □

Theorem 4.3 (Residual-comparison rate)**.** *Let $p > 2$. Suppose, at the computed states,*

$$\widehat{\delta}_m \ge c_U \left\| \widehat{G}_m - u^* \right\|_X^p , \quad c_U > 0, \tag{9}$$

$$S_m^X \ge \kappa_0 (m+1)^{-\chi} \left\| \mathcal{E}'(\widehat{G}_m) \right\|_{X^*} , \quad \kappa_0 > 0, \quad 0 \le \chi < \tfrac{1}{2}. \tag{10}$$

Assume (1)–(4), and, uniformly in m,

$$t_m^{\mathrm{pool}} \tau_m \ge \underline{t} > 0, \qquad (1 + t_m^{\mathrm{pool}}) \xi_m \le \vartheta t_m^{\mathrm{pool}} \tau_m S_m^X, \quad 0 \le \vartheta < 1.$$

With a_m as in (6), assume also $\varepsilon_m^{\mathrm{opt}} \le \zeta H_L(a_m)$, $0 \le \zeta < 1$. Then

$$\widehat{\delta}_m = O((m+1)^{-p(1-2\chi)/(p-2)}). \tag{11}$$

The constant depends on $p, c_U, \kappa_0, \underline{t}, L, \vartheta, \zeta$ and $\widehat{\delta}_0$. For the exact continuous method take $\tau_m = 1$, $\xi_m = \varepsilon_m^{\mathrm{opt}} = 0$.

Proof. Convexity and (9) give

$$\widehat{\delta}_m \le \left\| \mathcal{E}'(\widehat{G}_m) \right\|_{X^*} (\widehat{\delta}_m / c_U)^{1/p}.$$

For positive error this yields gradient domination. Thus $a_m \ge b_m^* := K(m+1)^{-\chi} \widehat{\delta}_m^{(p-1)/p}$, where $K = (1 - \vartheta) \underline{t} \kappa_0 c_U^{1/p}$. Descent and absorption imply monotonicity, so $b_m^* \le A := K \widehat{\delta}_0^{(p-1)/p}$. For $0 \le b \le A$, $H_L(b) \ge b^2 / (2 \max\{L, A\})$. Since H_L is increasing, (5) implies

$$\widehat{\delta}_{m+1} \le \widehat{\delta}_m - \frac{(1-\zeta) K^2}{2 \max\{L, A\}} (m+1)^{-2\chi} \widehat{\delta}_m^{2(p-1)/p}.$$

Apply Lemma 4.2. The case $\widehat{\delta}_0 = 0$ is immediate from descent and absorption. □

The norm comparison is imposed along the specified residual trajectory. The resulting energy exponent exceeds the atomic baseline exponent 1 when $\chi < 1/p$.

4.2 Energy and the natural distance

For the standard pure, regularized, and reaction p-energies, the energy–distance equivalence and the $W^{1,p}$ conversion are established in the energy–distance theorem and metric corollary in the main text. Therefore, if the energy estimate in Theorem 4.3 is $\widehat{\delta}_m = O(m^{-\alpha})$, then

$$d_{V,\bullet}(\widehat{G}_m, u^*) = O(m^{-\alpha/2}), \qquad \|\widehat{G}_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}),$$

under the corresponding zero-mean or reaction assumptions.

Thus (11) has natural-distance exponent $p(1-2\chi)/(2(p-2))$ and Sobolev exponent $(1-2\chi)/(p-2)$ for the pure, regularized, and reaction energies.

4.3 Local Hessian transfer

The transfer argument connects nonlinear greedy selection with Hilbert-space approximation. Xu–Xu [4, Theorem 3.1, equations (17)–(19)] analyze exact active-space minimization of a convex, L-smooth functional in a Sobolev Hilbert space. In their notation, the continuous selection strengths γ_i, source radius B, and innovations $\bar{g}_i = (I - P_{i-1})g_i$ yield

$$\delta_i \le \delta_{i-1} - \frac{\gamma_i^2 \delta_{i-1}^2}{2LB^2 \|\bar{g}_i\|^2}.$$

Their ReLU dictionary estimate then gives an energy rate $n^{-1-(2(k-s)+1)/d}$ for constant positive weakness in H^s, with $k \ge s$; strong convexity gives the corresponding norm bound. Here s denotes the Sobolev order. The argument below transfers nonlinear selection to a frozen metric through explicit local Hessian and relative-score conditions. The finite-pool estimate separates coverage, score evaluation, radial stationarity, and coefficient correction. The entropy argument applies the weak-selection volume estimate to the projected atoms with their inherited norms.

Let H be a real Hilbert space on which $\mathcal{E}$ is twice Fréchet differentiable in an open convex neighborhood of u^*. Assume $A_* = \mathcal{E}''(u^*)$ is bounded, symmetric, and coercive on H. Write $a_*(v,w) = \langle A_* v, w\rangle$ and $\|v\|_*^2 = a_*(v,v)$. Suppose the closed ball of radius R_* around u^* lies in that neighborhood and

$$|\langle [\mathcal{E}''(u) - A_*]v, w\rangle| \le L_* \|u-u^*\|_* \|v\|_* \|w\|_*, \qquad \|u-u^*\|_* \le R_*, \quad L_* R_* \le \tfrac{1}{2}. \tag{12}$$

For finite-dimensional $\Phi \subset H$ let z_Φ be the frozen orthogonal projection of u^* and $G_\Phi^\sharp$ an exact minimizer of $\mathcal{E}$ over Φ. Assume both points lie in this ball.

Lemma 4.4 (Taylor and projection estimates)**.** *Under* (12)*, for u in the ball,*

$$\left|\mathcal{E}(u) - \mathcal{E}(u^*) - \tfrac{1}{2}\|u-u^*\|_*^2\right| \le \tfrac{L_*}{6}\|u-u^*\|_*^3, \tag{13}$$

$$\left\|\mathcal{E}'(u) - A_*(u-u^*)\right\|_{*'} \le \tfrac{L_*}{2}\|u-u^*\|_*^2, \tag{14}$$

$$\left\|G_\Phi^\sharp - z_\Phi\right\|_* \le L_* \|z_\Phi - u^*\|_*^2. \tag{15}$$

Here the dual norm in (14) *is taken over H.*

Proof. One and two integrations of (12) give the first two estimates. Put $d = G_\Phi^\sharp - z_\Phi \in \Phi$. Use $\langle \mathcal{E}'(G_\Phi^\sharp), d\rangle = 0$ and $a_*(z_\Phi - u^*, d) = 0$. Subtracting $A_* d$ along the segment from z_Φ to $G_\Phi^\sharp$ bounds $\|d\|_*^2$ by $L_* R_* \|d\|_*^2 + (L_*/2)\|z_\Phi - u^*\|_*^2 \|d\|_*$. Absorb the first term. All segments remain in the ball by convexity. □

Assume $\mathcal{D}_X \subset H$ and the algorithmic dictionary satisfies the uniform norm comparison

$$0 < c_* \le \|g\|_* \le C_* < \infty \quad (g \in \mathcal{D}_X), \qquad \mathcal{D}_* = \{g/\|g\|_* : g \in \mathcal{D}_X\}. \tag{16}$$

Let Φ_m be nested finite-dimensional spaces with $\Phi_{m+1} = \Phi_m + \operatorname{span}\{g_{m+1}\}$, where each selected direction is independent of Φ_m. Denote their frozen orthogonal projections by P_m, and set $z_m = P_m u^*$, $r_m = \|u^* - z_m\|_*$, and $G_m^\sharp = G_{\Phi_m}^\sharp$. At an exact active-space state set

$$S_m^* = \sup_{d \in \mathcal{D}_*} |a_*(z_m - u^*, d)|, \qquad \eta_m^H = \sup_{g \in \mathcal{D}_X} |\langle \mathcal{E}'(G_m^\sharp), g\rangle - a_*(z_m - u^*, g)|.$$

The preceding lemma gives $\eta_m^H \le C_T \|z_m - u^*\|_*^2$, uniformly on the specified ball and dictionary. The relative-score condition below compares this defect with S_m^*.

Proposition 4.5 (Selection under a relative score condition)**.** *Suppose nonlinear selection over $\mathcal{D}_X$ has strength $t \in (0,1]$. For $0 < t_0 < tc_*/C_*$, the condition*

$$(1+t)\eta_m^H \le (tc_* - t_0 C_*) S_m^* \tag{17}$$

implies $|a_*(z_m - u^*, g_{m+1}/\|g_{m+1}\|_*)| \ge t_0 S_m^*$.

Proof. The selected unnormalized frozen score is at least $t \sup_{g \in \mathcal{D}_X} |a_*(z_m - u^*, g)| - (1+t)\eta_m^H$. The supremum is at least $c_* S_m^*$, and division by the selected norm costs at most C_*. This proves the assertion. □

Fix m_0, $Q_0 = I - P_{m_0}$ and

$$H_0 = Q_0 H, \qquad f_0 = Q_0 u^*, \qquad \mathcal{D}_{\mathrm{rst}} = Q_0 \mathcal{D}_* \setminus \{0\}.$$

The projected atoms retain their projected norms, which are at most one. Under the 2^{n-1}-ball convention, the 2^j-ball cover in the volume argument corresponds to ε_{j+1}.

Lemma 4.6 (Entropy bound for bounded-atom weak OGA)**.** *Let $\mathcal{D}_0$ be symmetric with atom norms at most one in a Hilbert space, $K = K_1(\mathcal{D}_0)$ compact, and $f_0 \in B_* K$ with $B_* \ge 0$. Let weak OGA of strength $t_0 > 0$ start from the zero space. After $j \ge 1$ independent extensions its error satisfies*

$$\|f_0 - P_j f_0\| \le \frac{B_*}{t_0} \frac{(j! V_j)^{1/j}}{\sqrt{j}} \varepsilon_{j+1}(K), \qquad V_j = \frac{\pi^{j/2}}{\Gamma(j/2+1)}. \tag{18}$$

Proof. If $B_* = 0$, then $f_0 = 0$ and the assertion is immediate. Suppose $B_* > 0$. Write $y_i = \|f_0 - P_i f_0\|^2$ and $l_i = \|(I - P_{i-1})d_i\| > 0$. Orthogonality and the source condition give the best score at least y_{i-1}/B_*. Hence $y_i \le y_{i-1} - t_0^2 y_{i-1}^2/(B_*^2 l_i^2)$. If the error reaches zero the result is immediate. Otherwise taking reciprocals and using the arithmetic–geometric mean inequality yields

$$y_j \le \frac{B_*^2}{t_0^2 \sum_{i=1}^j l_i^{-2}} \le \frac{B_*^2}{t_0^2 j} \Big(\prod_{i=1}^j l_i \Big)^{2/j}.$$

The cross-polytope generated by $\{\pm d_1, \dots, \pm d_j\}$ has j-dimensional volume $(2^j/j!) \prod_i l_i$ and lies in the orthogonal projection of K onto their span. A cover of K by 2^j balls of radius $\varepsilon_{j+1}(K) + h$, $h > 0$, projects to a cover of that cross-polytope. Volume comparison gives $\prod_i l_i \le j! V_j (\varepsilon_{j+1}(K) + h)^j$. Let $h \downarrow 0$ and substitute. □

The proof adapts the volume–entropy argument of Li–Siegel [2, Theorem 4.2 and equations (4.4)–(4.9)] to weak selection with atom norms bounded by one, retaining the innovation weights of the restarted spaces. Stirling's formula gives a uniform bound on the factorial–volume factor divided by $\sqrt{j}$.

Theorem 4.7 (Hessian transfer after restart)**.** *Assume* (12) *and* (16)*. For every state in* $m_0 \le m \le m_0 + J$*, suppose that the exact nonlinear state and frozen projection lie in the specified ball. For each selection step* $m_0 \le m < m_0 + J$*, assume nonlinear selection has strength at least* t*, and* (17) *holds with the same* $t_0 > 0$*. On the unrenormalized restarted dictionary assume*

$$f_0 \in B_* K_1(\mathcal{D}_{\mathrm{rst}}), \qquad \varepsilon_j(K_1(\mathcal{D}_{\mathrm{rst}}); H_0) \le C_{\mathrm{ent}} j^{-\beta}, \tag{19}$$

for $1 \le j \le J+1$*, with compact closed absolutely convex hull,* $B_*, C_{\mathrm{ent}} > 0$*, and* $\beta > 0$*. Then*

$$\|z_{m_0+j} - u^*\|_* \le C B_* C_{\mathrm{ent}} t_0^{-1} j^{-\beta}, \qquad \mathcal{E}(G^\sharp_{m_0+j}) - \mathcal{E}(u^*) \le C_{\mathrm{nl}} j^{-2\beta}.$$

Here C *is absolute and* C_{nl} *depends on* $L_*, R_*, B_*, C_{\mathrm{ent}}, t_0$*. The* 2^{n-1}*-ball convention gives the index* $j+1$ *in Lemma 4.6, so the entropy assumption is stated through* $J+1$*.*

Proof. Proposition 4.5 gives weak frozen selection. The residual is orthogonal to the starting space, so its scores against d and $Q_0 d$ agree. The spaces after the starting index therefore form weak OGA on H_0 with dictionary $\mathcal{D}_{\mathrm{rst}}$. Apply Lemma 4.6. Lemma 4.4 gives $\left\|G^\sharp_{m_0+j} - u^*\right\|_* \le r_{m_0+j} + L_* r_{m_0+j}^2 \le (1 + L_* R_*) r_{m_0+j}$, and (13) gives the energy bound. □

A family satisfying the structural assumptions. Take $H = H^1(0,1)$ and

$$\mathcal{E}_\lambda(u) = \int_0^1 \left(\tfrac{1}{2}|u'|^2 + \tfrac{1}{2}u^2 + \tfrac{\lambda}{4}u^4\right) \mathrm{d}x - \ell(u), \qquad \lambda \ge 0.$$

For a fixed integer $k \ge 2$ and $0 < a < b < 1$, put $g_t = (x-t)_+^k / \left\|(x-t)_+^k\right\|_{H^1}$, $t \in [a,b]$, and $\mathcal{D}_X = \{\pm g_t\}$. Choose any finite signed measure μ with total variation at most B_{src}, set $u^* = \int_a^b g_t \,\mathrm{d}\mu(t)$, and manufacture $\ell(v) = \int (u^{*\prime} v' + u^* v + \lambda (u^*)^3 v)$. The embedding $H^1(0,1) \hookrightarrow L^\infty$ makes the load continuous and the Hessian locally Lipschitz on H^1 balls. The variational load determines the natural boundary flux. Its bilinear form is $\int (v'w' + (1 + 3\lambda(u^*)^2) vw)$, which is coercive in H^1. The norm comparison (16) follows, the dictionary is compact, and $u^* \in C_* B_{\mathrm{src}} K_1(\mathcal{D}_*)$. Its closed absolutely convex hull is compact: finite nets for the dictionary approximate that hull by bounded finite-dimensional convex hulls. Projection by Q_0 preserves the source representation and bounds the projected entropy by the original entropy. These structural assumptions hold for the family of targets defined above. For $\lambda = 0$, $\eta_m^H = 0$ and the setting reduces to Hilbert OGA. For $\lambda > 0$, the relative-score condition is expressed through the following quantitative small-nonlinearity estimate. Let $C_{\mathrm{emb}} = \|H^1(0,1) \hookrightarrow L^\infty(0,1)\|$ (for the standard squared H^1 norm one may use the explicit bound $C_{\mathrm{emb}} \le \sqrt{2}$) and $M_* = \|u^*\|_{L^\infty}$, and use the H^1-normalized atoms above. Since a_* dominates the standard H^1 inner product, the quartic Hessian satisfies, on the specified R_*-ball,

$$L_\lambda := 3\lambda C_{\mathrm{emb}} (C_{\mathrm{emb}} R_* + 2M_*), \qquad L_\lambda R_* \le \tfrac{1}{2}, \tag{20}$$

and the norm-comparison and source constants can be chosen as

$$c_* = 1, \qquad C_\lambda = \sqrt{1 + 3\lambda M_*^2}, \qquad B_\lambda = C_\lambda B_{\mathrm{src}}. \tag{21}$$

Indeed, $\|g_t\|_* = \sqrt{1 + 3\lambda \|u^* g_t\|_{L^2}^2}$ lies between 1 and C_λ, and the change of variables from g_t to $g_t / \|g_t\|_*$ multiplies the atomic budget by at most C_λ. Put

$$C_{T,\lambda} = L_\lambda \left[1 + \tfrac{1}{2}(1 + L_\lambda R_*)^2\right]. \tag{22}$$

The projection estimates then give $\eta_m^H \le C_\lambda C_{T,\lambda} r_m^2$. Consequently the following explicit bound is sufficient for the relative-score condition:

$$\Delta_\lambda := (t c_* - t_0 C_\lambda) - (1+t) C_\lambda^2 C_{T,\lambda} B_{\mathrm{src}} > 0. \tag{23}$$

For fixed R_*, M_*, B_{src}, t, and $t_0 < t c_*$, continuity of the nonlinear term as $\lambda \downarrow 0$ gives an interval $0 < \lambda \le \lambda_0$ satisfying (20) and (23).

Sufficient conditions for the concrete semilinear family. Fix m_0, $Q_0 = I - P_{m_0}$, and the restarted dictionary $\mathcal{D}_{\mathrm{rst}} = Q_0\mathcal{D}_* \setminus \{0\}$, with the projected norms retained and $\mathcal{D}_* = \{g_t/\|g_t\|_* : t \in [a,b]\}$ including both signs. The sufficient conditions on $0 \le j \le J$ are:

(A1) an initial radius and a specified R_* for which every exact active minimizer, frozen projection, and corrected state in the stated finite range stays in $B_*(u^*, R_*)$; for a specified $r_{\mathrm{ent}} > 0$ this includes $r_{m_0} \le r_{\mathrm{ent}}$ and $L_\lambda R_* \le \frac{1}{2}$;

(A2) a uniform selection strength t and $t_0 < tc_*/C_\lambda$ for which the displayed bound Δ_λ is positive at every selection step $m_0 \le m < m_0 + J$ in the stated range;

(A3) a quantitative entropy estimate for this same projected, unrenormalized object, with the ball-counting convention fixed above,

$$\varepsilon_n(K_1(\mathcal{D}_{\mathrm{rst}}); H_0) \le C_{\mathrm{ent}} n^{-\beta_A}, \qquad 1 \le n \le J+1, \tag{24}$$

and $Q_0 u^* \in B_\lambda K_1(\mathcal{D}_{\mathrm{rst}})$; and

(A4) correction, quadrature, and radial errors absorbed in the finite-pool score bound, with tolerances prescribed from the available finite-pool quantities.

The exponent β_A and constant C_{ent} are those of the projected-dictionary entropy estimate (24).

Corollary 4.8 (Concrete-family rate)**.** *Assume the sufficient conditions above hold for some $\lambda \in (0, \lambda_0]$ and $0 < \beta_A$, and use exact active-space minimization in the statement of the rate. Then the restarted CGA trajectory satisfies, for $1 \le j \le J$,*

$$\|z_{m_0+j} - u^*\|_* \le C B_\lambda C_{\mathrm{ent}} t_0^{-1} j^{-\beta_A}, \qquad \mathcal{E}_\lambda(G^\sharp_{m_0+j}) - \mathcal{E}_\lambda(u^*) \le C_{\mathrm{nl}} j^{-2\beta_A}. \tag{25}$$

The constants are those of Theorem 4.7.

Function-space setting. The concrete semilinear family uses the compact parameter interval $[a,b] \subset (0,1)$. The sinh model uses an L^∞-bounded neighborhood for local Hessian estimates and finite-dimensional spaces for the trajectory analysis. Pure p-energies are treated through the global energy geometry and the projection bounds below.

4.4 Finite-reference scores and restricted stationarity

On a specified finite-dimensional space Y_M with a positive frozen inner product, assume $u^* \in Y_M$ and $\widehat{\Phi}_m \subset Y_M$. In this setting let z_m be the frozen projection of u^* onto $\widehat{\Phi}_m$. Let the fixed reference set $\mathcal{A}_M \subset Y_M$ contain all pools on the stated finite range. Assume $0 < c_M \le \kappa_g := \|g\|_* \le C_M < \infty$. Set

$$h_m(g) = |a_*(u^* - z_m, g)|, \quad S_m^{\mathrm{val}} = \max_{g \in \mathcal{A}_M} h_m(g)/\kappa_g, \quad \tau_m^{\mathrm{val}} = \frac{\max_{g \in \mathcal{P}_m} h_m(g)/\kappa_g}{S_m^{\mathrm{val}}}$$

when $S_m^{\mathrm{val}} > 0$. If the computed scores obey $|\widetilde{s}_m(g) - h_m(g)| \le \Delta_m$ on $\mathcal{P}_m$, the proof of Proposition 4.5 gives

$$\frac{h_m(g_{m+1})}{\kappa_{g_{m+1}}} \ge \frac{t_m^{\mathrm{pool}} c_M \tau_m^{\mathrm{val}} S_m^{\mathrm{val}} - (1 + t_m^{\mathrm{pool}})\Delta_m}{C_M}. \tag{26}$$

Thus the following nonnegative lower bound $(t_m^{\mathrm{pool}} c_M \tau_m^{\mathrm{val}} - t_0 C_M) S_m^{\mathrm{val}} - (1 + t_m^{\mathrm{pool}})\Delta_m$ suffices for weakness t_0 relative to this finite reference set. The additional comparison $S_m^{\mathrm{val}} \ge (1 - \zeta_M) S_m^*$,

with $0 \le \zeta_M < 1$, transfers the finite-reference estimate to the continuous score. The numerical records report quadrature sensitivity through the comparison of two quadrature rules.

The active-space KKT residual in a reference norm $\|\cdot\|_{0,M}$ is

$$k_m = \sup_{\substack{v \in \widehat{\Phi}_m \\ \|v\|_{0,M}=1}} |\langle \mathcal{E}'(\widehat{G}_m), v\rangle|. \tag{27}$$

The supremum is over the active space, even when its norm is inherited from Y_M; set $k_m = 0$ if $\widehat{\Phi}_m = \{0\}$. If $\mathcal{E}$ is $\mu_{\mathrm{act}} > 0$ strongly convex on the active space and its exact minimizer exists, then

$$\left\|\widehat{G}_m - G_m^\sharp\right\|_{0,M} \le k_m/\mu_{\mathrm{act}}, \quad \mathcal{E}(\widehat{G}_m) - \mathcal{E}(G_m^\sharp) \le k_m^2/(2\mu_{\mathrm{act}}).$$

The first follows by strong monotonicity paired with the state difference; the second follows by minimizing the strong-convexity quadratic. A bound C_{score} on $\left\|\mathcal{E}''(u)g|_{\widehat{\Phi}_m}\right\|_{\widehat{\Phi}_m^*}$ over the relevant segments and pool gives an additional score error at most $C_{\mathrm{score}}k_m/\mu_{\mathrm{act}}$. The coercivity constants depend on the chosen reference norm and trial space. Since $\widehat{\Phi}_m \subset Y_M$,

$$\inf_{0\neq v\in\widehat{\Phi}_m} \frac{a_*(v,v)}{\|v\|_{0,M}^2} \;\ge\; \inf_{0\neq v\in Y_M} \frac{a_*(v,v)}{\|v\|_{0,M}^2}.$$

Thus the coercivity constant is associated with the space over which the Rayleigh quotient is minimized.

4.5 An approximate-source estimate for the computed trajectory

Let Y be a finite-dimensional real Hilbert space containing u^*, all active spaces, and a fixed finite symmetric dictionary $\mathcal{D}_{\mathrm{fin}}$ of frozen-unit vectors. For $m_0 \le m < m_0 + J$ assume

$$\widehat{\Phi}_{m+1} = \widehat{\Phi}_m + \mathrm{span}\{d_{m+1}\}, \quad d_{m+1} \in \mathcal{D}_{\mathrm{fin}}, \quad d_{m+1} \notin \widehat{\Phi}_m. \tag{28}$$

The dictionary must contain every accepted direction in this finite range. Here P_m denotes the frozen orthogonal projection onto $\widehat{\Phi}_m$. Set

$$\begin{aligned} z_m = P_m u^*, \qquad e_m = u^* - z_m, \qquad r_m = \|e_m\|_*, \\ S_m^{\mathrm{fin}} = \max_{d\in\mathcal{D}_{\mathrm{fin}}} |a_*(e_m, d)|, \qquad \ell_m = \|(I - P_m)d_{m+1}\|_* > 0. \end{aligned}$$

Define $\theta_m = |a_*(e_m, d_{m+1})|/S_m^{\mathrm{fin}}$ if $S_m^{\mathrm{fin}} > 0$, and $\theta_m = 0$ otherwise. The vector $(I - P_m)d_{m+1}$ is the innovation of the candidate d_{m+1}, that is, its component orthogonal to $\widehat{\Phi}_m$, and ℓ_m is the norm of that innovation. Let $Q_0 = I - P_{m_0}$ and assume explicit coefficients satisfy

$$v_B = \sum_{d\in\mathcal{D}_{\mathrm{fin}}} a_d Q_0 d, \quad \sum_d |a_d| \le B_{\mathrm{entry}}, \quad \|Q_0 u^* - v_B\|_* \le \sigma, \qquad B_{\mathrm{entry}} \ge 0. \tag{29}$$

The vectors $Q_0 d$ retain their projected norms. The coefficient budget B_{entry} belongs to this approximate restarted representation, while B_{src} denotes the continuous atomic source budget.

Theorem 4.9 (Weighted finite-trajectory source bound)**.** *Under* (28)–(29)*, take* $0 \le \underline{\theta}_m \le \theta_m$ *and set*

$$W_j = \sum_{m=m_0}^{m_0+j-1} \frac{\underline{\theta}_m^2}{\ell_m^2}, \qquad x_0 = (r_{m_0}^2 - \sigma^2)_+.$$

If $B_{\mathrm{entry}} > 0$ and $x_0 > 0$, then for $0 \le j \le J$,

$$(r_{m_0+j}^2 - \sigma^2)_+ \le \left(x_0^{-1} + \frac{W_j}{4B_{\mathrm{entry}}^2} \right)^{-1}. \tag{30}$$

In particular, if $W_j > 0$, $r_{m_0+j}^2 \le \sigma^2 + 4B_{\mathrm{entry}}^2/W_j$. If $x_0 = 0$ or $B_{\mathrm{entry}} = 0$, all later errors are at most σ. For $W_j = 0$, projection monotonicity gives the bound.

Proof. For $w_m = (I - P_m)d_{m+1}$, Pythagoras gives

$$r_m^2 - r_{m+1}^2 = \frac{|a_*(e_m, d_{m+1})|^2}{\ell_m^2}. \tag{31}$$

The residual is orthogonal to both the starting and current spaces. Consequently

$$r_m^2 = a_*(e_m, Q_0 u^*) \le B_{\mathrm{entry}} S_m^{\mathrm{fin}} + \sigma r_m, \qquad S_m^{\mathrm{fin}} \ge \frac{(r_m^2 - \sigma^2)_+}{2B_{\mathrm{entry}}}$$

when $B_{\mathrm{entry}} > 0$; the last step uses $2(r_m^2 - \sigma r_m) \ge r_m^2 - \sigma^2$. While both successive errors exceed σ, put $x_m = r_m^2 - \sigma^2 > 0$. Equation (31) implies $x_{m+1} \le x_m - b_m x_m^2$ with $b_m = \underline{\theta}_m^2/(4B_{\mathrm{entry}}^2 \ell_m^2)$. Taking reciprocals proves (30) up to a crossing. After a crossing, monotonicity makes the positive part zero. If $S_m^{\mathrm{fin}} = 0$ or $B_{\mathrm{entry}} = 0$, the source pairing gives $r_m^2 \le \sigma r_m$ directly. This also covers zero errors. □

Barron–Cohen–Dahmen–DeVore prove in [1, Theorem 2.3, equation (2.31)] that exact OGA in a Hilbert space satisfies

$$\|f - f_N\|_H^2 \le \|f - h\|_H^2 + \frac{4\,\|h\|_{L^1(\mathcal{D})}^2}{N}, \qquad h \in L^1(\mathcal{D}). \tag{32}$$

Here $L^1(\mathcal{D})$ denotes the atomic variation space. In the restarted setting, $f = Q_0 u^*$, $h = v_B$, $\|f - h\|_* \le \sigma$, and the coefficient budget on the projected dictionary is at most B_{entry}. The source pairing and reciprocal estimate use this classical mechanism, with Theorem 4.9 retaining the observed weights $\underline{\theta}_m^2/\ell_m^2$ along the generated spaces. If $\underline{\theta}_m \ge \underline{\theta} > 0$, then $\ell_m \le 1$ gives $W_j \ge j\underline{\theta}^2$. The upper bound decreases as the accumulated weight W_j grows.

Li–Siegel [2, Corollary 4.2 and equation (4.11)] obtain the approximate-source inequality

$$\frac{1}{2}\big(\|r\|^2 - \|f - h\|^2\big) \le \|h\|_{\mathcal{A}}\, |\langle r, g\rangle|,$$

which leads to the same innovation-weighted projection recursion. The present theorem specializes this mechanism to the restarted, unrenormalized projected dictionary and retains the observed $\underline{\theta}_m^2/\ell_m^2$ weights generated by the nonlinear selection rule.

4.6 An exact quartic energy connection

For the one-dimensional pure or regularized quartic energy define

$$X = \{u \in W^{1,4}(0,1) : \textstyle\int_0^1 u = 0\}, \qquad \mathcal{E}_\varepsilon(u) = \textstyle\int_0^1 \left(\frac{\varepsilon^2}{2}|u'|^2 + \frac{1}{4}|u'|^4 \right) \mathrm{d}x - \ell(u).$$

An additive constant in the density is immaterial. Assume $u^* \in X$ solves the weak equation and let $Y \subset X$ contain u^* and the spaces/dictionary of Section 4.5. Use

$$a_*(v, w) = \int_0^1 (\varepsilon^2 + 3|(u^*)'|^2) v' w' \,\mathrm{d}x.$$

For $\varepsilon > 0$ this is positive definite on Y. For $\varepsilon = 0$ assume explicitly that $\int |(u^*)'|^2 |v'|^2 > 0$ for every nonzero $v \in Y$; the sufficient condition $(u^*)' \neq 0$ almost everywhere ensures this on the zero-mean space. Finiteness follows from Hölder's inequality.

For any $\widehat{G}_m \in \widehat{\Phi}_m$, define

$$D_m = \left\| \widehat{G}_m - z_m \right\|_*, \quad v_m = \widehat{G}'_m - (u^*)', \quad R_m = \int_0^1 \left((u^*)' v_m^3 + \tfrac{1}{4} v_m^4 \right) \mathrm{d}x.$$

Proposition 4.10 (Quartic identity and comparison)**.** *For every such state,*

$$\widehat{\delta}_m = \tfrac{1}{2}(r_m^2 + D_m^2) + R_m. \tag{33}$$

If $D_m \leq q_{\mathrm{corr}} r_m$ and $|R_m| \leq \rho(r_m^2 + D_m^2)$ throughout a finite range, where $q_{\mathrm{corr}} \geq 0$ and $0 \leq \rho < 1/2$, then

$$(\tfrac{1}{2} - \rho) r_m^2 \leq \widehat{\delta}_m \leq C_E r_m^2, \qquad C_E = (\tfrac{1}{2} + \rho)(1 + q_{\mathrm{corr}}^2). \tag{34}$$

If U_m is a source upper bound for r_m^2, the additive estimate is $\widehat{\delta}_m \leq U_m/2 + D_m^2/2 + |R_m|$.

Proof. Set $a = (u^*)'$ and $v = v_m$. Subtracting the linear part of the density at a gives $\varepsilon^2 v^2/2 + 3a^2 v^2/2 + av^3 + v^4/4$. The weak equation cancels the integrated linear part, including the load. Also $\widehat{G}_m - z_m$ lies in the active space and $u^* - z_m$ is orthogonal to it, so $\left\| \widehat{G}_m - u^* \right\|_*^2 = D_m^2 + r_m^2$. This proves the identity; the bounds follow by substitution. □

In (33), D_m and R_m account for the nonlinear correction at the computed state. The numerical records report the quadrature defect separately. The manufactured solution determines the values of q_{corr}, ρ, and the source remainder used in the finite-trajectory comparisons.

4.7 Projection bounds under a power-type decrease condition

Retain the projections of Section 4.5 and set

$$x_m = r_m^2, \qquad \gamma_m = \frac{|a_*(e_m, d_{m+1})|}{\ell_m}.$$

Then (31) is $x_{m+1} = x_m - \gamma_m^2$. When $S_m^{\mathrm{fin}} > 0$, $\gamma_m = \theta_m S_m^{\mathrm{fin}} / \ell_m$; otherwise $\gamma_m = 0$ by its direct definition. The power-type condition below relates the normalized innovation decrease to the projection residual.

Theorem 4.11 (Explicit power-type finite-range bound)**.** *Let $\alpha > 0$, $c_0 > 0$, and assume, at every step with $r_m > 0$ in $m_0 \leq m < m_0 + J$,*

$$\gamma_m^2 \geq c_0 r_m^{2+2/\alpha}. \tag{35}$$

For $r_{m_0} > 0$ and $0 \leq j \leq J$,

$$r_{m_0+j}^2 \leq \left(r_{m_0}^{-2/\alpha} + \frac{c_0 j}{\alpha} \right)^{-\alpha}. \tag{36}$$

If $r_{m_0} = 0$, every later projection error is zero. For the quartic models, if the relative conditions of Proposition 4.10 also hold, then

$$\widehat{\delta}_{m_0+j} \leq C_E \left(r_{m_0}^{-2/\alpha} + \frac{c_0 j}{\alpha} \right)^{-\alpha}. \tag{37}$$

Natural-distance and Sobolev bounds are obtained by taking respectively the one-half and one-quarter powers, with the constants of the energy–distance theorem in the main text.

Proof. For $x_{m+1} > 0$, convexity gives

$$x_{m+1}^{-1/\alpha} - x_m^{-1/\alpha} \geq \frac{x_m - x_{m+1}}{\alpha x_m^{1+1/\alpha}} \geq \frac{c_0}{\alpha}.$$

Summation proves (36). If a residual reaches zero, nestedness gives the remaining cases. Substitute the projection bound into (34), then apply the energy–distance theorem in the main text. □

A sufficient factorization on positive-error steps is

$$S_m^{\mathrm{fin}} \geq c_1 r_m^{2\nu}, \qquad \theta_m/\ell_m \geq c_2, \qquad \nu = \tfrac{1}{2} + \tfrac{1}{2\alpha}, \qquad c_1, c_2 > 0. \tag{38}$$

Indeed $\gamma_m^2 \geq (c_1 c_2)^2 r_m^{4\nu}$ and $4\nu = 2 + 2/\alpha$. Thus $c_0 = (c_1 c_2)^2$ is admissible. The residual visibility factor and the selected-direction factor are the two factors in this lower bound.

For a finite sequence with strictly positive drops, every prescribed $\alpha > 0$ admits

$$c_{\min}(\alpha; J) = \min_{\substack{m_0 \leq m < m_0 + J \\ x_m > 0}} \frac{x_m - x_{m+1}}{x_m^{1+1/\alpha}} > 0.$$

This constant defines an a posteriori envelope on the specified sequence. Continuation evaluates the same inequality with the calibration constants held fixed. The numerical records give the explicit bounds, constants, and index ranges.

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